# Between-sample quality control and exploratory diagnostics

## Groups

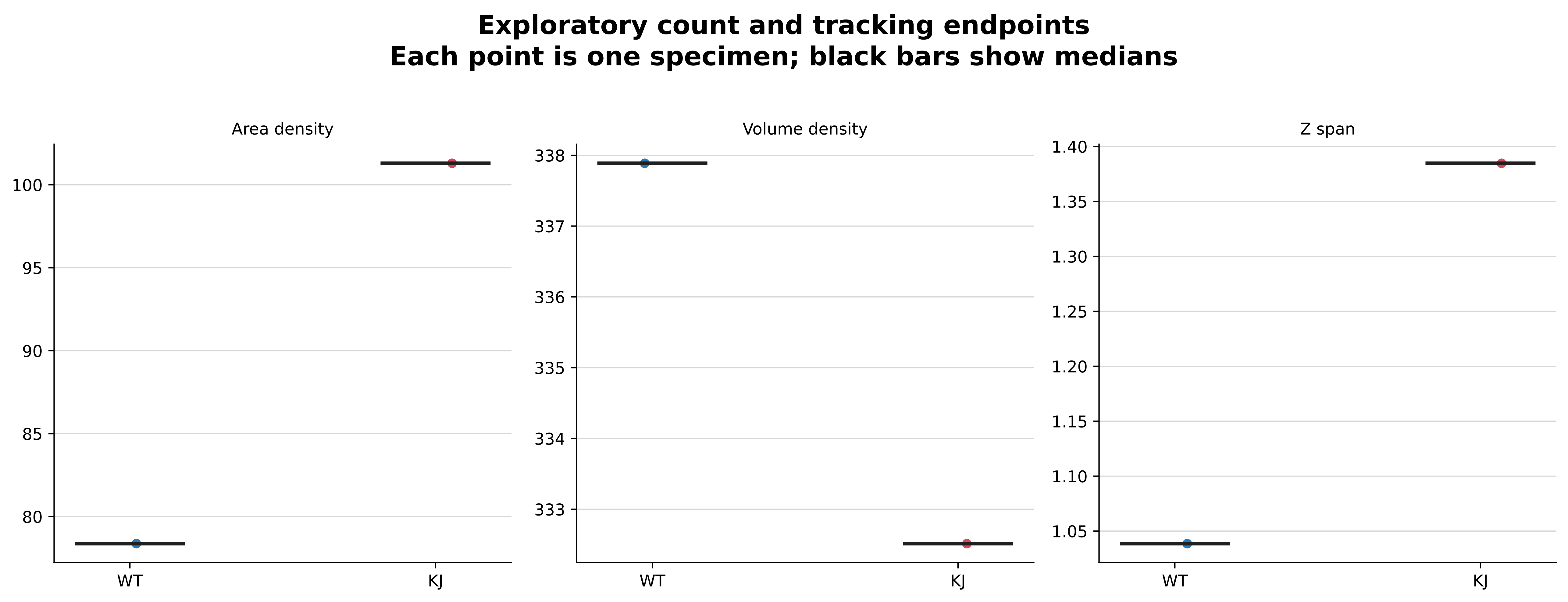
Reference: WT. Comparison: KJ.

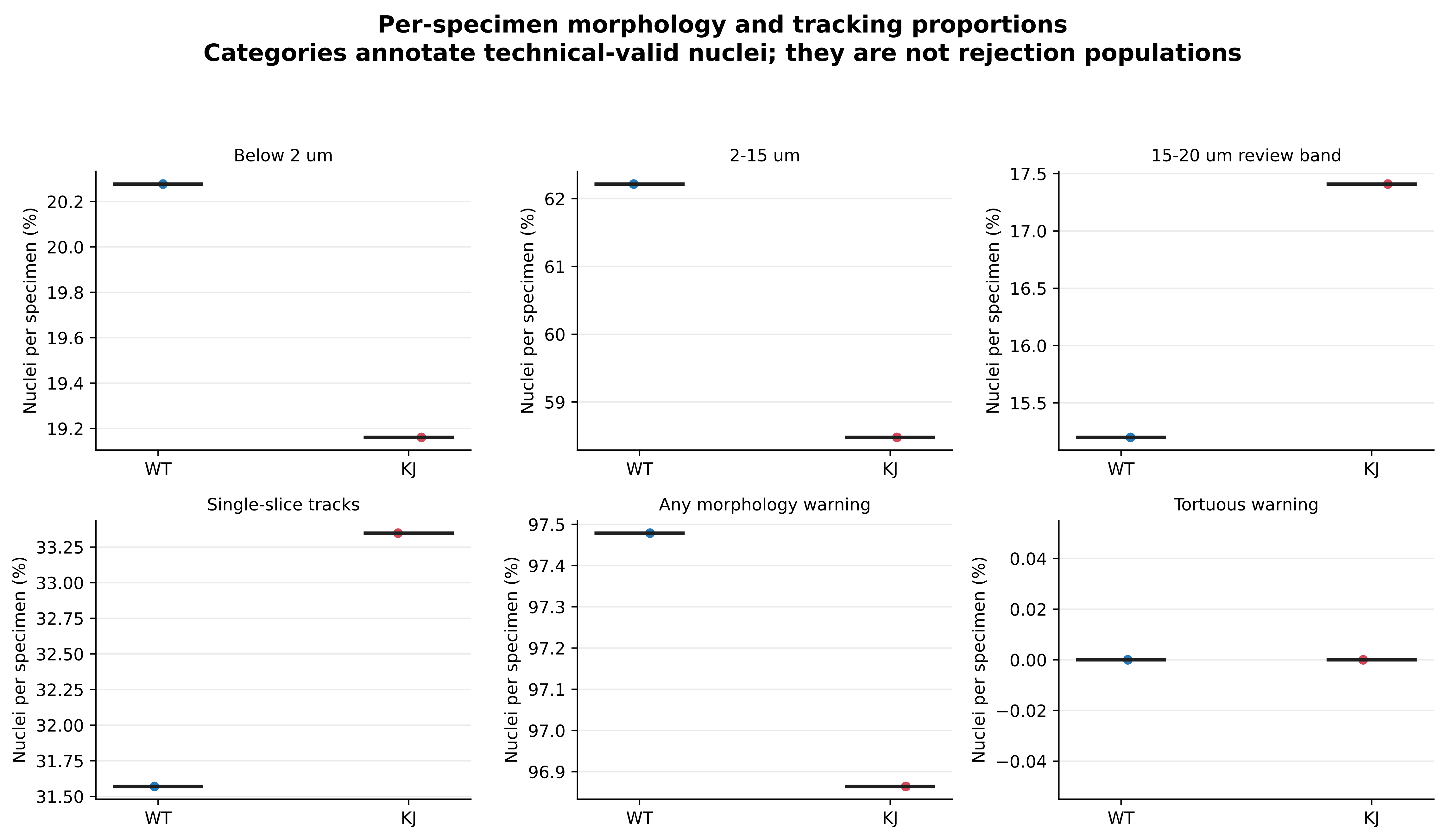
## Purpose

Use this package to judge acquisition comparability, normalization, tracking behavior, and warning frequencies.

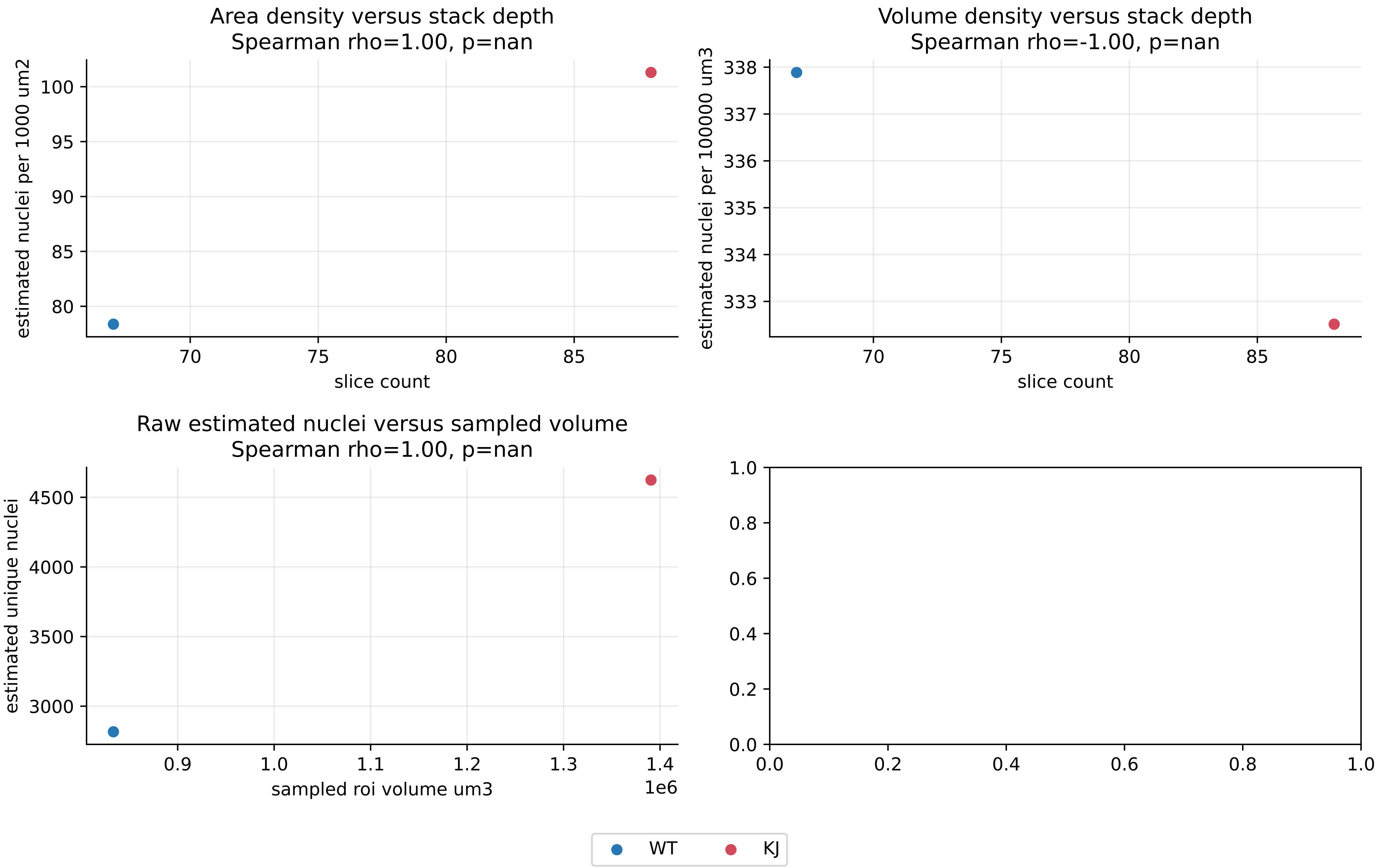
## Primary results

Biological conclusions belong in ../01_biological_results.


Exploratory count and tracking endpoints
Each point is one specimen; black bars show medians
Area density
100
95
90
85
80
WT
KJ
Volume density
338
337
336
335
334
333
WT
KJ
Z span
1.40
1.35
1.30
1.25
1.20
1.15
1.10
1.05
WT
KJ


Per-specimen morphology and tracking proportions
Categories annotate technical-valid nuclei; they are not rejection populations
Below 2 um
2-15 um
15-20 um review band
Single-slice tracks
Any morphology warning
Tortuous warning
Nuclei per specimen (%)
WT
KJ
20.2
20.0
19.8
19.6
19.4
19.2
62
61
60
59
17.5
17.0
16.5
16.0
15.5
33.25
33.00
32.75
32.50
32.25
32.00
31.75
31.50
97.5
97.4
97.3
97.2
97.1
97.0
96.9
0.04
0.02
0.00
−0.02
−0.04

**Count and acquisition-depth diagnostics**
**Count endpoints remain exploratory**



## Specimen-level statistical summary
## Each value is calculated across biological specimens, not pooled nuclei

| Measure | WT median | KJ median | Median change (%) | Effect size Cliff's delta | Median test FDR q | Rank test FDR q | Mean test FDR q |
|---|---|---|---|---|---|---|---|
| Estimated nuclei per 1,000 um2 | 78.4 | 101 | +29.3% | +nan | nan | nan | nan |
| Estimated nuclei per 100,000 um3 | 338 | 333 | -1.6% | +nan | nan | nan | nan |
| Specimen median Z span (um) | 1.04 | 1.38 | +33.3% | +nan | nan | nan | nan |

Medians: the typical specimen value in each group.  Median change: (KJ median - WT median) / WT median x 100; positive means higher in KJ.
Cliff's delta: direction and separation, from -1 to +1; positive means KJ specimens tend to have higher values.  FDR q: p-value adjusted for testing multiple measurements; q < 0.05 is evidence of a group difference.
Median test: permutation comparison of specimen medians (primary).  Rank test: Mann-Whitney comparison of specimen ordering.  Mean test: Welch comparison of specimen means.

# How to use the quality-control package

## Not the primary biological result

These panels assess acquisition depth, count normalization, tracking span, morphology-warning frequencies, and processing provenance. They should not replace the biological morphology report.

## Count endpoints

Area- and volume-normalized counts remain exploratory when they retain an association with slice count or sampled volume. The fixed-depth sensitivity panel helps assess this dependence.

## Morphology warnings

Warnings annotate unusual but technical-valid nuclei. They are not rejection populations and must not be used to force either group toward reference morphology.

## Tracking Z span

Z span is sensitive to acquisition spacing and cross-slice linking. A single-slice nucleus can still be biologically genuine.